

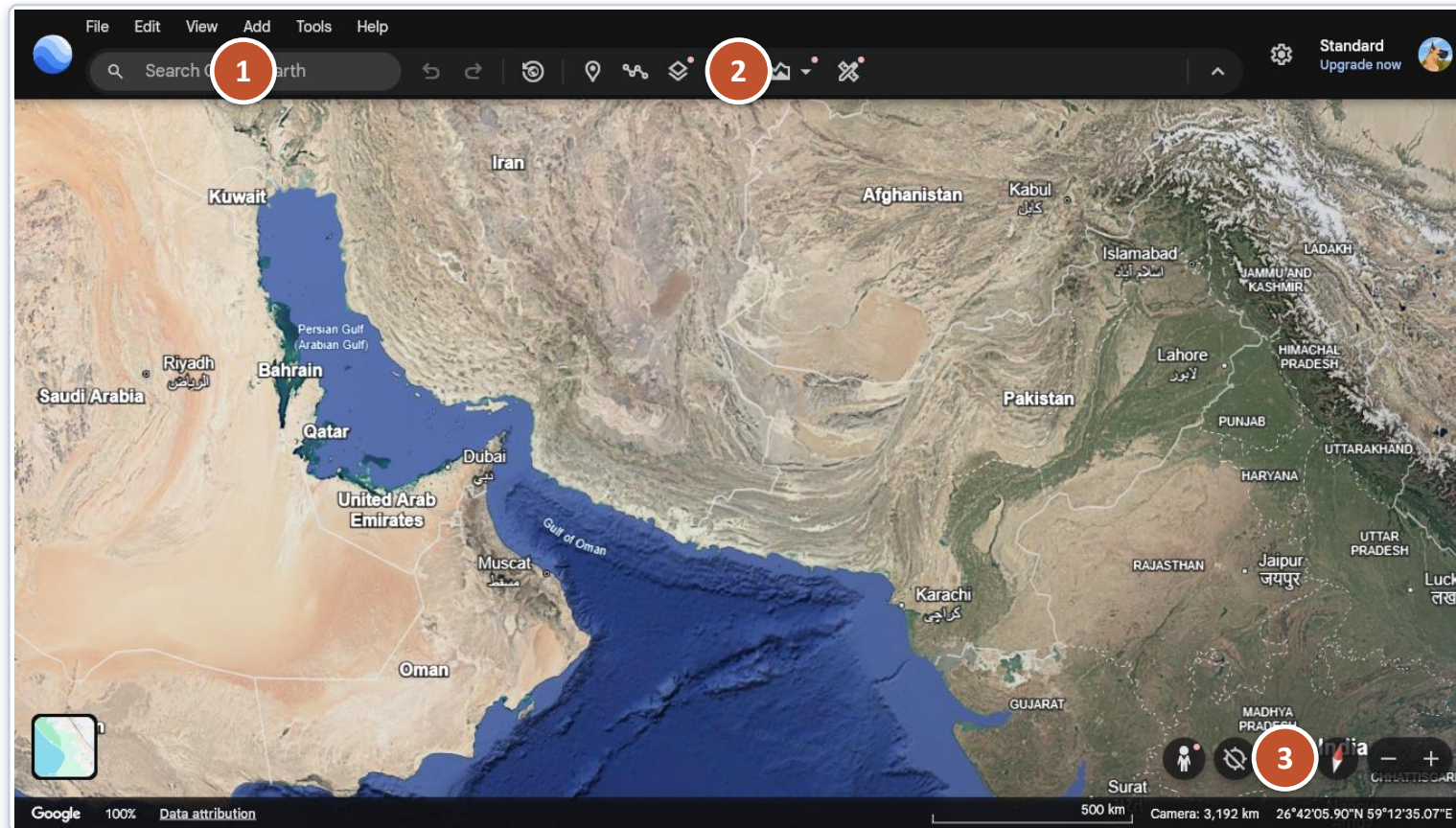
HCI – CG

Assignment Task - 1

By: ABDUL HAYEE {2K24/CSE/4}

HCI Focus: Interaction Elements

App analyzed: Google Earth (web) — satellite view of South Asia / Middle East



Callouts 1–3 are marked directly on the screenshot and match the numbers at right.

1 Search Bar

Lets the user type a place name or address and jump straight to it. This is the primary wayfinding affordance — it turns an open-ended “where is X?” question into a single-step action.

2 Measure / Tool Icons (top bar)

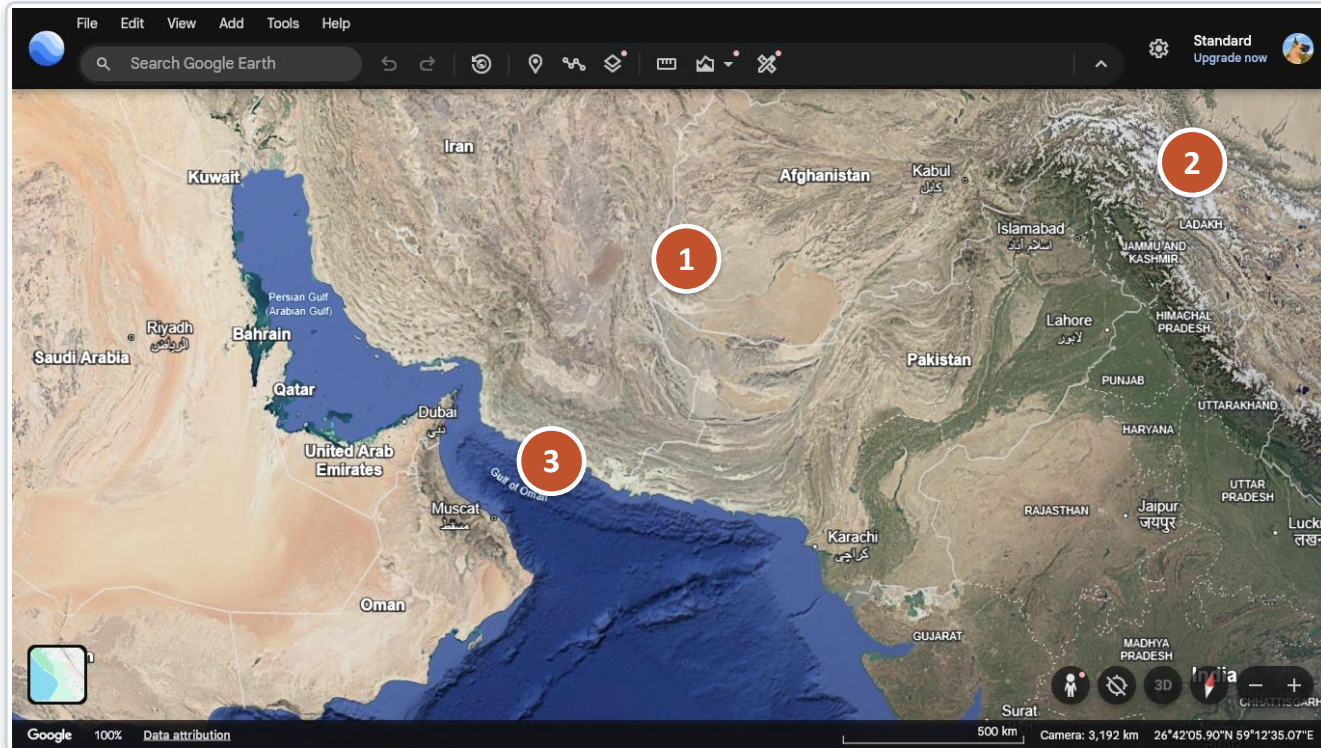
The ruler and related icons expose specific commands (measure distance, add placemark, view layers) as always-visible buttons, so users can discover and trigger features without hunting through menus.

3 3D / Navigation Control Cluster

The tilt, compass, and zoom buttons in the bottom-right corner give direct, incremental control over the camera, letting users reorient and move through the 3D globe with immediate visual feedback.

CG Focus: Rendering Elements

Same screenshot — viewed for graphics rendering rather than controls



1

Photorealistic Texture Mapping

Real satellite/aerial imagery is draped over the terrain mesh as a texture, giving every mountain range and desert its true surface color and pattern instead of a flat, generic fill.

2

Relief Shading on 3D Terrain Mesh

Snow-capped peaks and ridgelines (e.g., near the Himalayas) show light and shadow gradients across the underlying elevation mesh, letting a flat image communicate real mountainous height.

3

Depth Cues via Color Gradation (Bathymetry)

Ocean color shifts from pale near the coast to deep blue further out, a shading convention that stands in for actual water depth and reinforces the 3D shape of the seafloor.

Reflection Questions

1. Does CG quality help or hinder interaction? The photorealistic imagery and relief shading make it much easier to recognize real places and judge terrain at a glance, which helps users navigate confidently. But loading high-resolution satellite texture over a large area can lag on a slow connection, briefly freezing the view mid-interaction.

2. One simple HCI improvement? Add short hover labels (tooltips) to the top toolbar icons, since several icons (measure, layers, imagery date) look similar and give no clue to a first-time user about what each one does.